Kernix 0.9.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

25 workloads

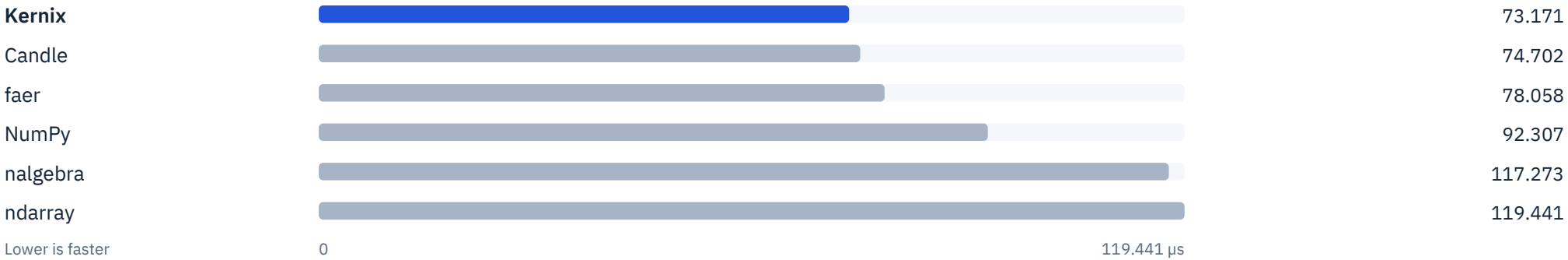
100 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

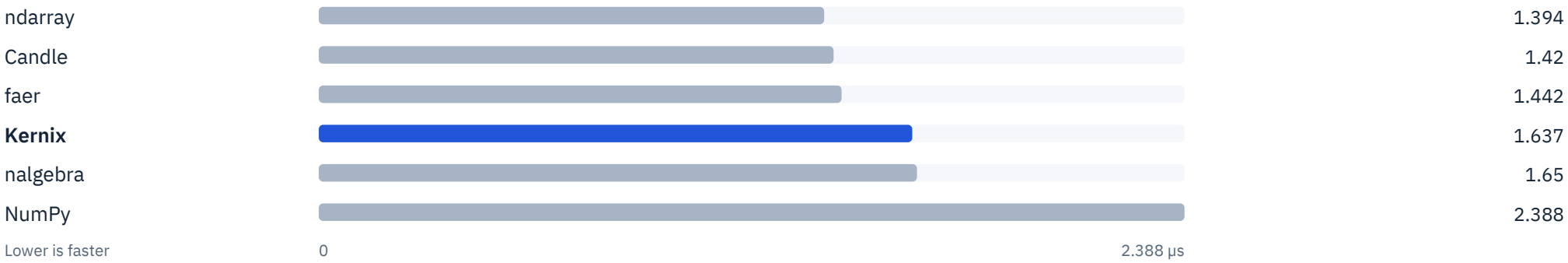
Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

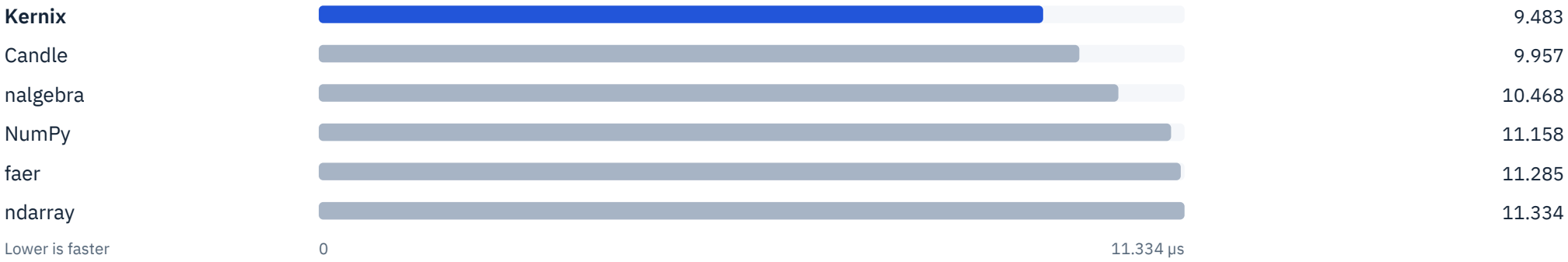
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



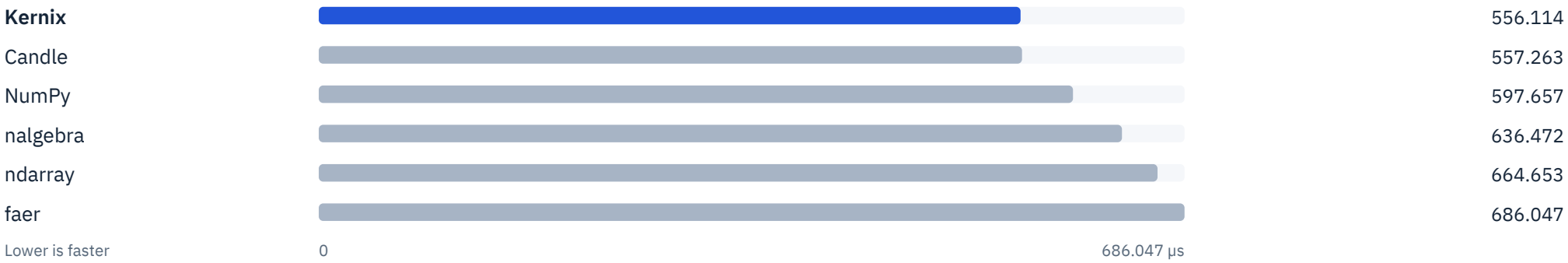
Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



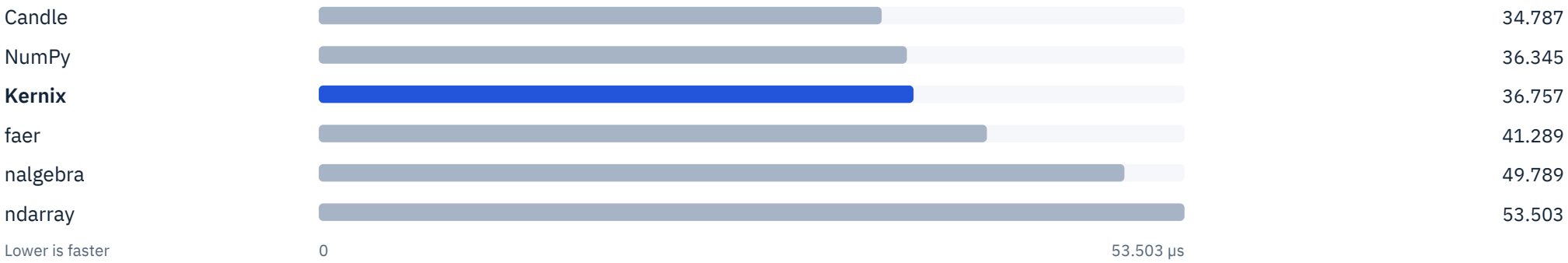
Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

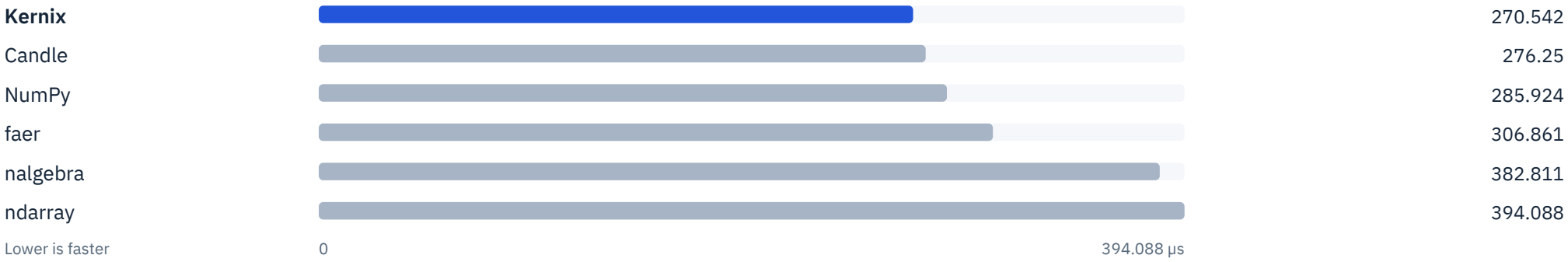
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

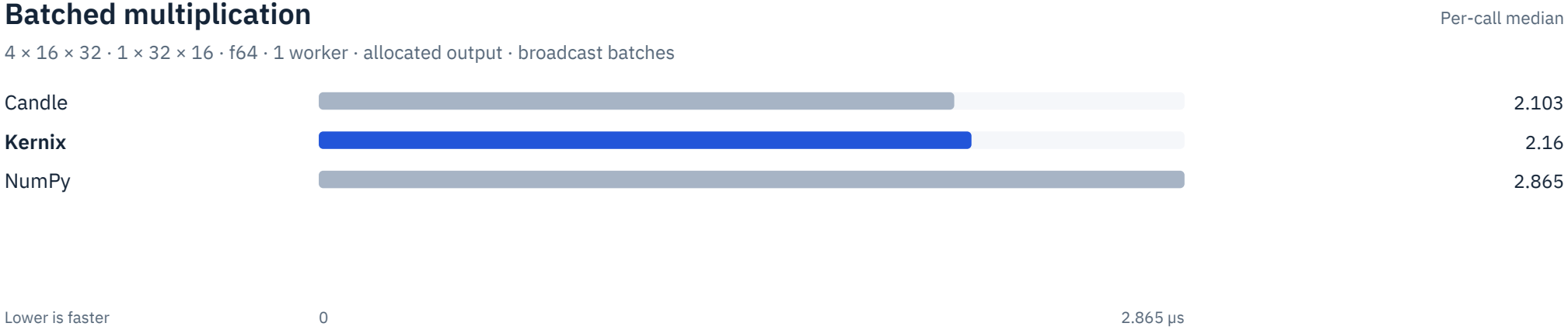


Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

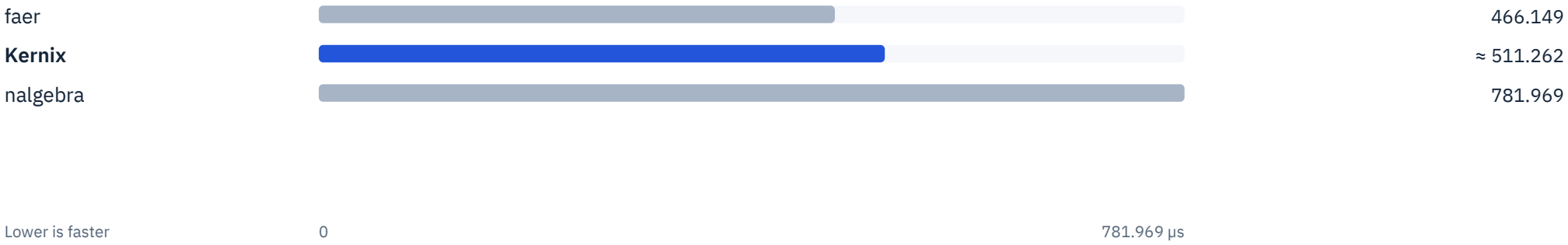
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

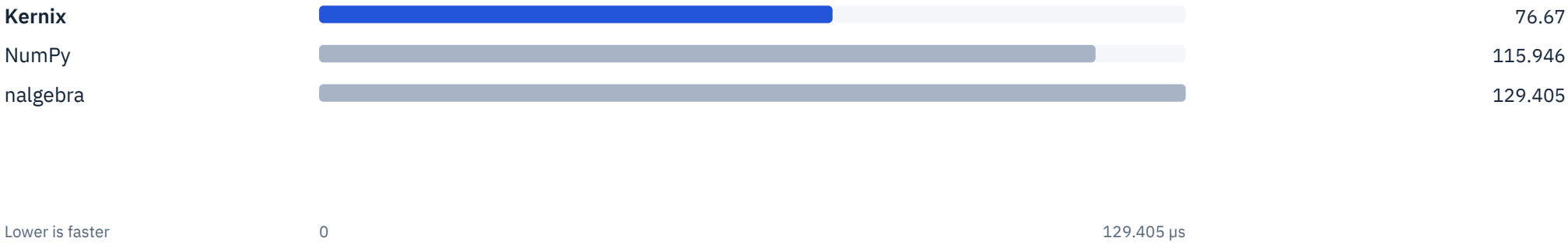
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

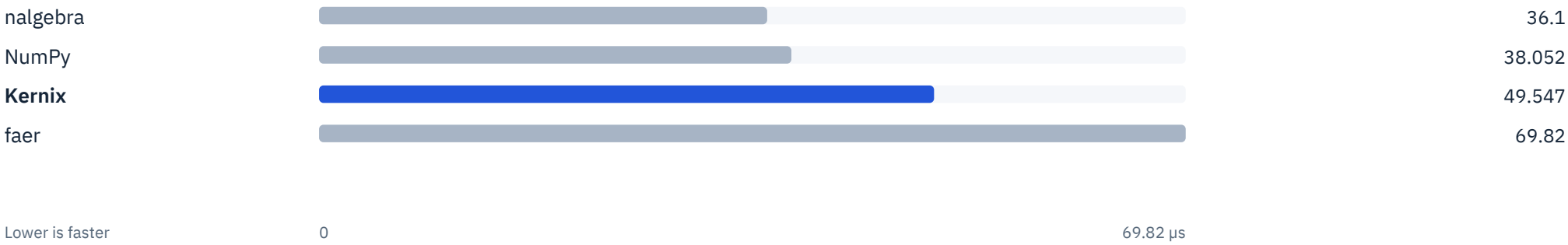


Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

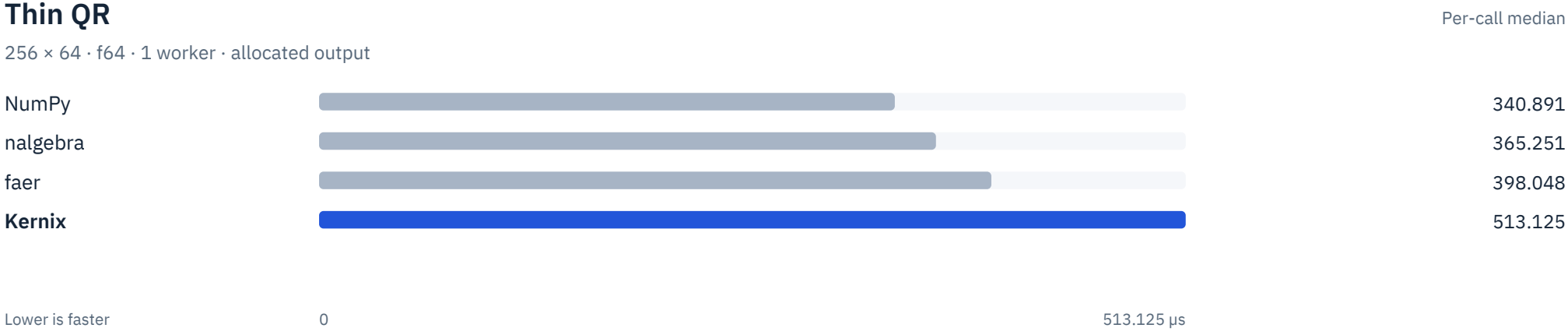
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



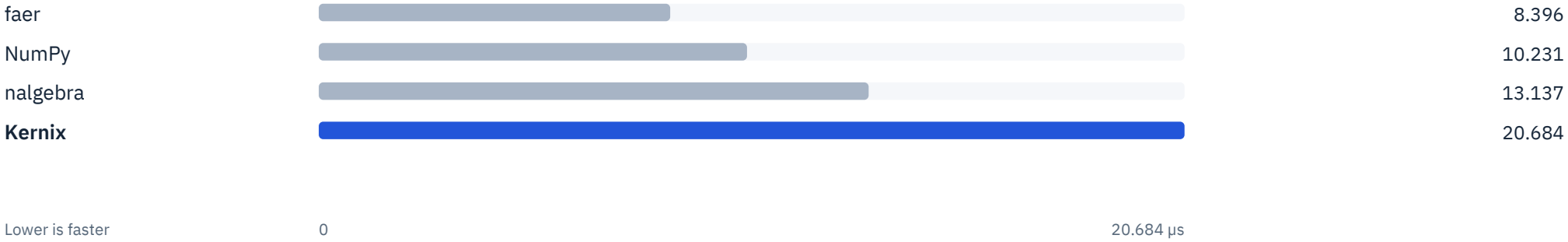
Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

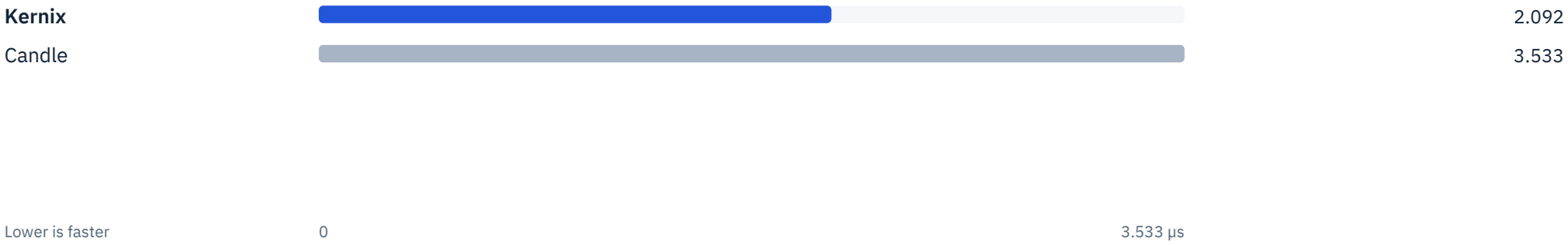
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

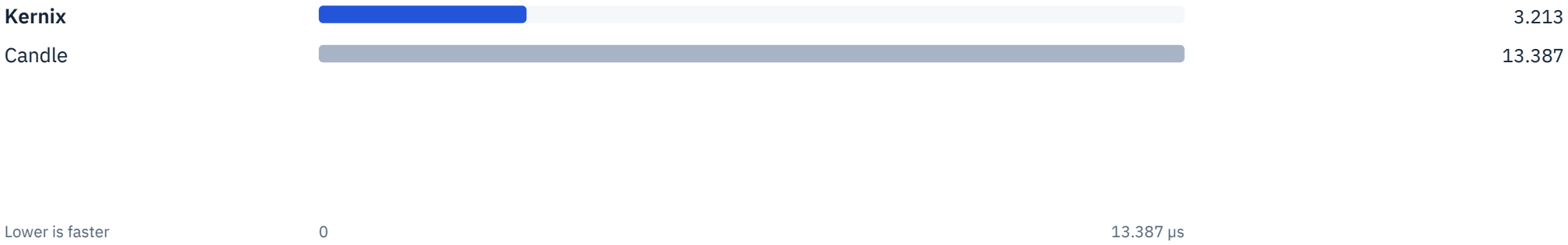
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

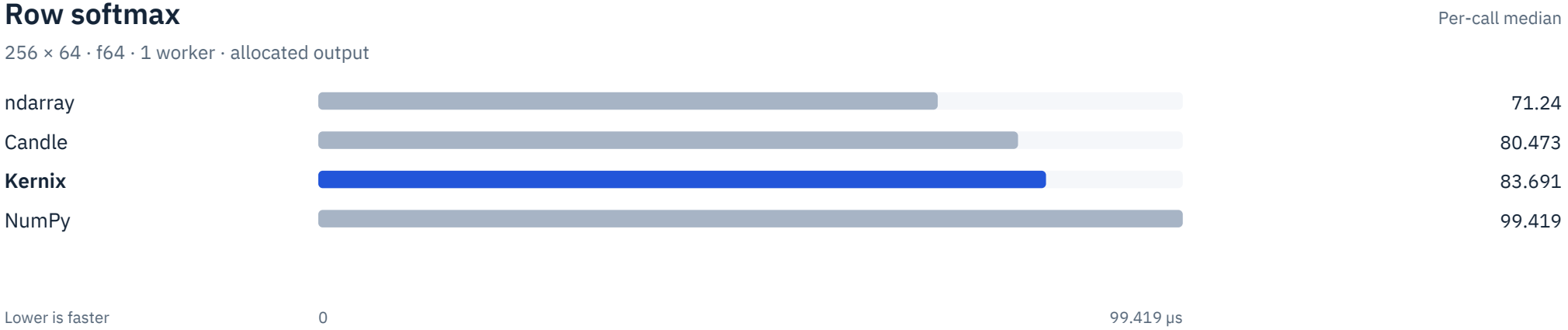


Kernix 0.9.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



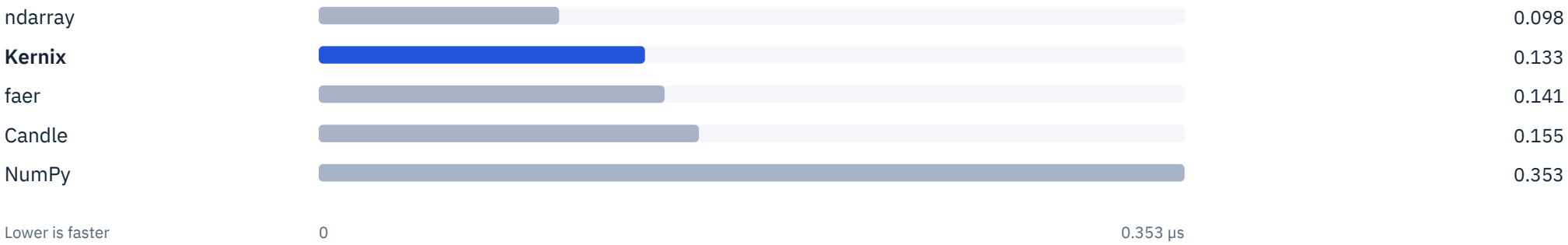
Kernix 0.9.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

64 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9773944919635305	0.9235327313769752 – 1.0403771163433535

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9789425417092872	0.9268104333538715 – 1.018511826247222

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0069935500152878	0.9932964002130146 – 1.025894234196159

Thin QR

96 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0491846686820097	0.9656443002198263 – 1.0593508293468055

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.020154379165327	0.9954990521073648 – 1.0457764236184073

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0018462734428142	0.9800260353103897 – 1.0123633757391148

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	**0.9867207984437114**	95% INTERVAL	0.9328305380091483 – 1.0138829376981746

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1637	1.637
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1442	1.442
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1394	1.394
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1420	1.42
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2388	2.388
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1650	1.65
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11158	11.158
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	10468	10.468
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9483	9.483
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11334	11.334

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11285	11.285
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9957	9.957
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	78058	78.058
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	92307	92.307
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	117273	117.273
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	119441	119.441
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	73171	73.171
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	74702	74.702
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	636472	636.472
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	597657	597.657

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	556114	556.114
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	557263	557.263
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	686047	686.047
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	664653	664.653
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	5642	5.642
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6276	6.276
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4885	4.885
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5213	5.213
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5708	5.708
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7242	7.242

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	36757	36.757
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	36345	36.345
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49789	49.789
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	34787	34.787
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	53503	53.503
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	41289	41.289
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	285924	285.924
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	306861	306.861
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	270542	270.542
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	382811	382.811

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	276250	276.25
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	394088	394.088
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2103	2.103
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2865	2.865
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2160	2.16
Column-pivoted QR / faer 96 × 32 · f64	71206	71.206
Column-pivoted QR / Kernix 96 × 32 · f64	56637	56.637
Column-pivoted QR / nalgebra 96 × 32 · f64	78306	78.306
Column-pivoted QR / nalgebra 256 × 64 · f64	781969	781.969
Column-pivoted QR / faer 256 × 64 · f64	466149	466.149

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	511261.99999999994	511.26199999999994
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	49903	49.903
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	63224	63.224
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	71614	71.614
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	76670	76.67
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	115946	115.946
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	129405	129.405
Thin QR / faer 96 × 32 · f64	69820	69.82
Thin QR / NumPy 96 × 32 · f64	38052	38.052
Thin QR / nalgebra 96 × 32 · f64	36100	36.1

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	49547	49.547
Thin QR / faer 256 × 64 · f64	398048	398.048
Thin QR / Kernix 256 × 64 · f64	513125	513.125
Thin QR / nalgebra 256 × 64 · f64	365251	365.251
Thin QR / NumPy 256 × 64 · f64	340891	340.891
Cholesky factorization / Kernix 64 × 64 · f64	20684	20.684
Cholesky factorization / nalgebra 64 × 64 · f64	13137	13.137
Cholesky factorization / faer 64 × 64 · f64	8396	8.396
Cholesky factorization / NumPy 64 × 64 · f64	10231	10.231
Cholesky factorization / faer 128 × 128 · f64	37323	37.323

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	122242.99999999999	122.24299999999999
Cholesky factorization / NumPy 128 × 128 · f64	68058	68.058
Cholesky factorization / nalgebra 128 × 128 · f64	96829	96.829
Cholesky factorization / nalgebra 256 × 256 · f64	639843	639.843
Cholesky factorization / faer 256 × 256 · f64	191370	191.37
Cholesky factorization / Kernix 256 × 256 · f64	619867	619.867
Cholesky factorization / NumPy 256 × 256 · f64	510375	510.375
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2366	2.366
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4268	4.268
Model inference / Kernix batched normalized/sliced MLP · f64	2092	2.092

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3533	3.533
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2656	2.656
Model recording + forward / Candle batched normalized/sliced MLP · f64	5141	5.141
Model backward / Candle prepared graph → fresh gradients · f64	13387	13.387
Model backward / Kernix prepared graph → fresh gradients · f64	3213	3.213
Parameter update / Kernix transactional SGD, fixed gradients · f64	552	0.552
Row softmax / ndarray 256 × 64 · f64	71240	71.24
Row softmax / NumPy 256 × 64 · f64	99419	99.419
Row softmax / Candle 256 × 64 · f64	80473	80.473
Row softmax / Kernix 256 × 64 · f64	83691	83.691

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	133	0.133
Create zeros / faer 64 × 64 · f32	141	0.141
Create zeros / ndarray 64 × 64 · f32	98	0.098
Create zeros / NumPy 64 × 64 · f32	353	0.353
Create zeros / Candle 64 × 64 · f32	155	0.155
Create ones / faer 64 × 64 · f32	184	0.184
Create ones / Kernix 64 × 64 · f32	203	0.203
Create ones / Candle 64 × 64 · f32	237	0.237
Create ones / NumPy 64 × 64 · f32	1348	1.348
Create ones / ndarray 64 × 64 · f32	154	0.154